

EECS 836 Machine Learning Project: Health Prediction on MIMIC-III Dataset

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1 Overview

This project uses the MIMIC-III ICU electronic health record (EHR) dataset to build and evaluate machine learning models for clinical prediction. MIMIC-III contains deidentified hospital and ICU data (demographics, diagnoses, procedures, medications, labs/vitals, notes, outcomes) from a large tertiary care center [2, 1].

Team project requirement. This project must be completed in groups. You are required to form a team of 3 to 4 students to complete this assignment.

You must complete all three prediction tasks below. All students must use the same data splits, preprocessing rules, and evaluation metrics. You will implement a provided baseline and then improve it using **one** of three method tracks.

2 Data Access and Compliance (PhysioNet)

MIMIC-III is credentialed data hosted on PhysioNet [1]. Each student must obtain **individual** access.

2.1 How to Access MIMIC-III

1. Create a PhysioNet account and complete the **credentialing** process.
2. Complete required human-subject/data-privacy training via the CITI “**Data or Specimens Only Research**” (or current PhysioNet-specified equivalent), then submit the training report through PhysioNet [4, 3].
3. Sign the PhysioNet **Credentialed Health Data Use Agreement (DUA)** for MIMIC-III [5].
4. After approval, download the dataset from the official MIMIC-III page [1].
5. Do Not Share Data or Access. PhysioNet explicitly requires **each user** to obtain their own access rights and indicates that sharing within teams/classes is not permitted [6].

Data Privacy Warning: Process all sensitive data exclusively on local machines or approved university servers. Do not expose patient-level data by uploading it to publicly accessible repositories.

3 Library: PyHealth

Please use **PyHealth** to avoid building EHR parsing and common modeling pipelines from scratch. PyHealth provides dataset loaders (e.g., `MIMIC3Dataset`), preprocessing utilities, and example training scripts [7]. For practical guidance, students should follow the official PyHealth tutorials page [8]. You are still responsible for complying with PhysioNet access rules and the MIMIC-III DUA [1, 5].

3.1 Tasks

You must complete the following three tasks. You may share a common backbone and train three task-specific heads, or train separate models per task. If you use PyHealth, follow the official tutorials page [8] for end-to-end examples and best practices. To ensure results are comparable and reproducible, set the random seed to a fixed value at the start of every run. In particular, call PyHealth’s `set_seed(seed)` utility from `pyhealth/utis.py` [9] with `seed=42`. Your final report must include baseline results and your method’s improvement Δ over baseline for all three tasks.

3.1.1 Task 1: Drug Recommendation (Multi-label)

Problem: For each admission (visit) t of a patient, predict the set of medications administered/prescribed in that *current* admission t (e.g., ATC codes), using the patient’s cumulative clinical context.

Input: Use diagnoses and procedures from admissions up to and including time t (i.e., history $0..t$), and use *only past* medication history from admissions $0..t-1$ (i.e., exclude current-visit drugs from the input). This matches the PyHealth task design where `drugs_hist` excludes the current visit’s drugs.

Label space: Use a fixed medication vocabulary (e.g., top- N most frequent ATC codes). Document your vocabulary construction and any filtering rules.

Metrics (required):

- **Sample-wise AUPRC** (AUPRC-sample): compute AUPRC per admission-example and average across examples.
- **Precision@k** and **Recall@k** (e.g., $k \in \{10, 20\}$).

PyHealth support: See the PyHealth tutorials [8] and use the following baseline example as a starting point: `drug_recommendation_mimic3_transformer.py` [12].

3.1.2 Task 2: In-Hospital Mortality Prediction (Binary)

Problem: Predict whether the patient dies during the hospital stay.

Input window: Use only the first W hours of data (staff will set W , e.g., 24 or 48 hours) from the admission time.

Metrics (required): AUROC and AUPRC.

PyHealth support: See the PyHealth tutorials [8] and use the following baseline example as a starting point: `mortality_mimic3_rnn.py` [10].

3.1.3 Task 3: Hospital Readmission Prediction (Binary)

Problem: Predict whether the patient will be readmitted within D days after discharge.

Definition: Course staff will specify the readmission horizon D (commonly 30 days) and exclusion rules (e.g., planned readmissions if applicable).

Inputs: Use information available before discharge (or within a fixed early window if staff specifies).

Metrics (required): AUROC and AUPRC.

PyHealth support: See the PyHealth tutorials [8] and use the following baseline example as a starting point: `readmission_mimic3_rnn.py` [11].

4 Methods

You will choose **one** of the following three method categories and apply it consistently across **all three tasks**. Your submission must include at least one ablation demonstrating that the added component improves performance beyond the baseline.

4.1 Category 1: Advanced Deep Learning Methods

Goal: Improve predictive performance using stronger architectures for sequential EHR modeling (e.g., attention/transformers), temporal learning, or more expressive encoders.

Suggested directions:

1. **Attention/Transformer-based EHR modeling:** Implement an attention-based sequence model (e.g., reverse-time attention as in RETAIN) or a Transformer encoder for visit sequences/event sets, and compare against the baseline RNN while reporting both accuracy gains and efficiency trade-offs [13, 15, 16].
2. **Graph/Hypergraph deep models (as “advanced DL”):** Model higher-order dependencies across codes and visits using graph/hypergraph neural architectures [17, 18, 19, 20].

Minimum deliverables: (1) clear architecture description; (2) at least one ablation removing the added DL component; (3) training/inference cost discussion.

4.2 Category 2: Knowledge Graph (KG) / Ontology-Based Methods

Goal: Inject structured medical knowledge (code hierarchies, groupings, or curated relations) to improve generalization and clinical consistency.

Suggested directions:

1. **Ontology attention / ancestor aggregation:** Represent codes using ontology ancestors or attention over ancestor paths [21, 22, 24].
2. **Multiple ontology representations:** Use multiple hierarchical views of medical concepts (e.g., different ancestor sets/relations) to build richer code embeddings [23, 25].

3. **KG-augmented medication recommendation:** Incorporate medical knowledge paths or KG edges to guide medication prediction [26, 27, 28, 29, 30].

Minimum deliverables: (1) graph/ontology construction details; (2) ablation without KG/ontology component; (3) analysis showing where KG helps/hurts.

4.3 Category 3: LLM-Based Methods (Local/Offline Only)

Goal: Use language-model-driven priors or retrieval to improve representations/prediction, while respecting data access policies (i.e., do **not** send credentialed data to online services).

Suggested directions:

1. **Retrieval augmentation for EHR prediction:** Retrieve task-relevant medical context and fuse it with structured EHR encoders [31, 32, 33].
2. **Graph + auxiliary text / semantics:** Use auxiliary clinical text or code descriptions jointly with graph/temporal learning [34, 35].

Minimum deliverables: (1) strict reproducibility (fixed `seed=42`); (2) ablation removing the LLM/retrieval component; (3) compute/memory impact; (4) compliance statement.

5 Deliverables

5.1 Code (Reproducible)

- One-command training for baseline and your method (configs included).
- Fixed seeds; logs; checkpointing; metric scripts.

5.2 Report (6–10 pages)

Latex template: <https://www.overleaf.com/latex/templates/neurips-2024/tpsbbbrdqcmsh>

1. Task definitions and cohort construction
2. Baseline description and results
3. Your method track with design choices
4. Main results table (baseline vs method) for all tasks
5. Ablations
6. Optional error analysis (case studies)
7. Optional efficiency (time, memory, hardware)
8. Limitations and compliance statement

5.3 Presentation (8–10 minutes)

Motivation → method → results → takeaways.

6 Grading Rubric (110 points)

1) Correctness & Reproducibility (25 points)

- End-to-end runnable pipeline
- No data leakage
- Reproducibility practices (10)

2) Method Quality (25 points)

- Technical soundness
- Justified design choices

3) Evidence & Analysis (20 points)

- **Baseline comparison:** Includes a baseline-vs-method table and reports Δ improvement (at minimum on the required metrics).
- **Ablation study:** At least one meaningful ablation removing the added component(s) to show contribution.

4) Report Clarity (15 points)

5) Presentation (15 points)

6) Extra Credit (up to 10 points)

Choose one or more of the following (max 10 total):

- **Robustness to missingness/sparsity :** Stress tests with dropped features/visits or missingness-aware evaluation.
- **Interpretability:** Clinically meaningful explanations (attention/feature attribution), error case studies, or model auditing.

7 Suggested Reporting Tables (Include in Report)

Your report must include a **Baseline Comparison** table and an **Ablation Study** table. For the baseline comparison, report the improvement Δ for **all three tasks**. [?]

7.1 Baseline Comparison (Required)

7.2 Ablation Study (Required)

Model	Task 1: Drug Rec.		Task 2: Mortality		Task 3: Readmission	
	AUPRC	Precision@k	AUPRC	AUROC	AUPRC	AUROC
Baseline						
Your Method						
Δ Improvement						

Table 1: Baseline comparison for all three tasks. Report Δ as (Your Method – Baseline).

Model Variant	Task 1: Drug Rec.		Task 2: Mortality		Task 3: Readmission	
	AUPRC	Precision@k	AUPRC	AUROC	AUPRC	AUROC
Your Method (Full)						
Full (w/o Component 1)						
Full (w/o Component 2)						

Table 2: Ablation study. “Component” should correspond to a major added module (e.g., KG module, retrieval, auxiliary loss).

8 References

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